CME Natural Gas Options: Executive Summary

2019-01-02 to 2022-12-30 | Walk-Forward Backtest | No Look-Ahead

Objective

Develop and backtest two options strategies on CME Henry Hub Natural Gas (NG) futures:

(1) Strangle alpha overlay — sell short-dated strangles when vol is elevated and about to revert.

(2) Vol-targeted defensive allocation — exposure scaled by target_vol/rv20, with crisis-onset protective puts.

All signals use only past data. Models are re-calibrated on rolling windows. No future information is used.

Dataset	Size	Period / Type	Details
Futures	1,008 trading days	2019-01-02 to 2022-12-30	OI-weighted front month
Options	247,405 observations	Calls + Puts	Strike, DTE, IV, Greeks
Frequency	Daily	252 days/year	No intraday data

Key Findings from Exploratory Data Analysis

F1 Fat tails: kurtosis = 1.84. 31× more 4-sigma events than normal. BS underprices OTM options.

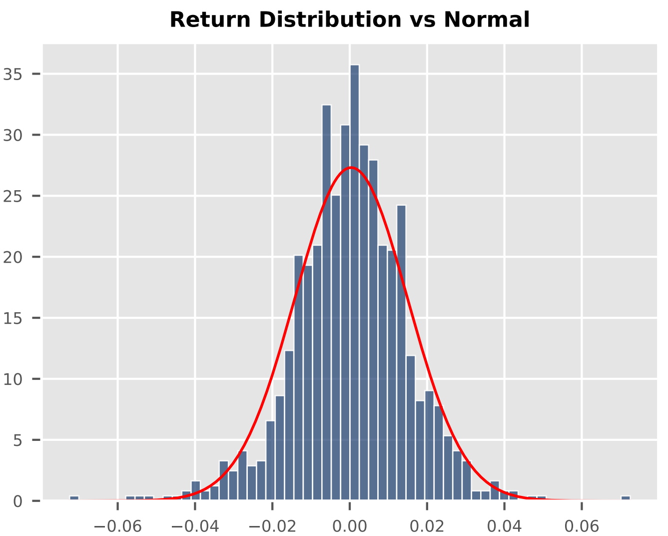
F2 Vol clusters and reverts: ARCH test $p < 1e\text{-}9$. Vol persists 1–2 weeks, mean-reverts in 1–2 months.

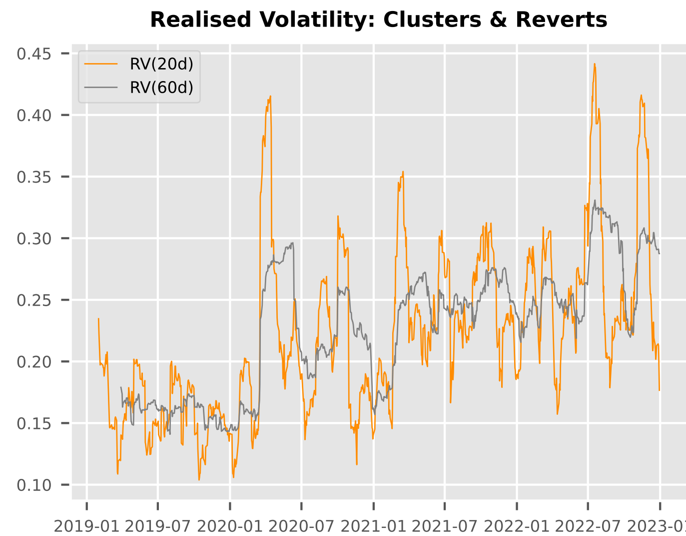
F3 Negative VRP: IV < RV on 100% of days. Opposite of equities — naive short-vol fails.

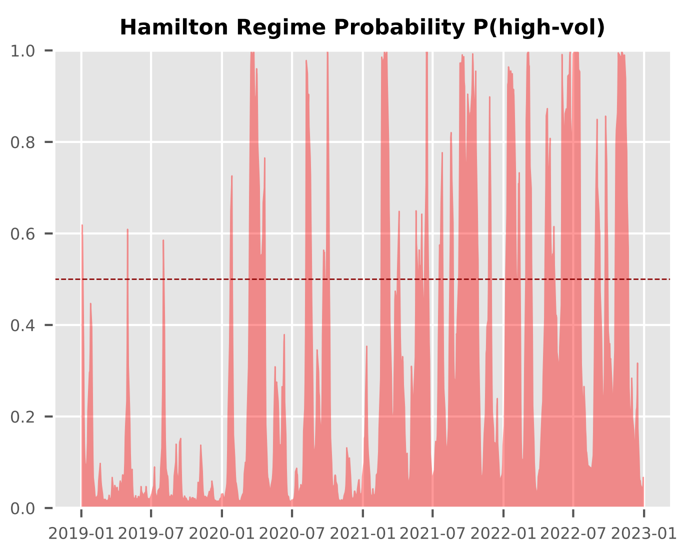
F4 Two regimes: Hamilton MS identifies low-vol (17.3%) and high-vol (32.4%) states.

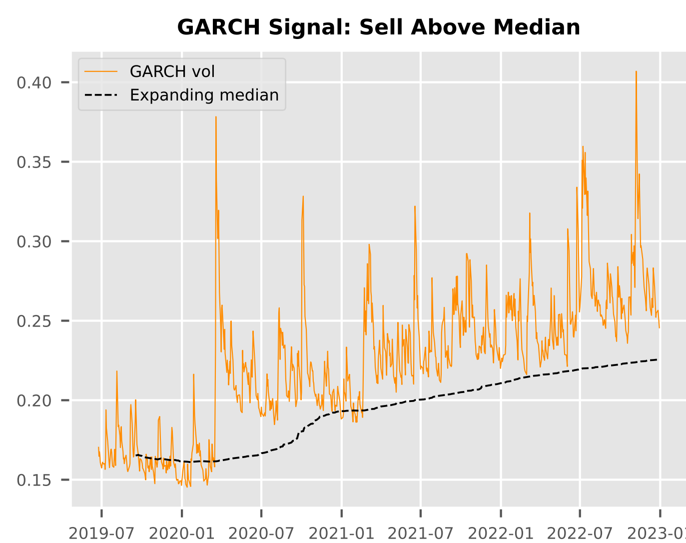
F5 Seasonality: Dec lowest vol (16%), Mar highest (28%). Thursday +8% (EIA storage report).

F6 Jump clustering: Poisson dispersion = 1.31. Months with 2+ jumps hurt short strangles.









Strategy Descriptions

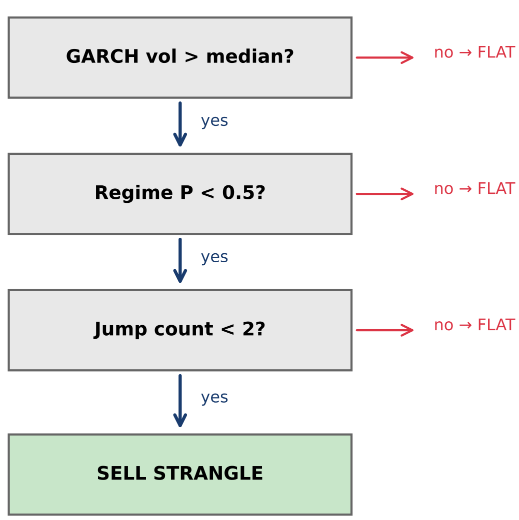
```
Strategy 1: Strangle Alpha Overlay
```

```
Structure:  Overlay on long futures base. Portfolio = B&H + strangle alpha.
Instrument: Short strangle (sell 1 OTM call + 1 OTM put) on NG futures.
Strikes:    25-delta (K_call = S·exp(+σ√T·Φ⁻¹(0.75)), K_put = S·exp(−σ√T·Φ⁻¹(0.75))).
Maturity:   21 calendar days. Hold to expiry. Stop-loss at 5.0× premium.
Sizing:     Up to 10 concurrent positions. Min 2-day gap between entries.
Pricing:    Black-Scholes with RV(20d) as vol input. Min premium $0.20.

Entry signal — ALL must hold simultaneously:
  Filter 1: EGARCH(1,1) vol > expanding median   →  vol elevated, theta income is rich
  Filter 2: Hamilton regime P(high-vol) < 0.5     →  market calm, gamma risk is low
  Filter 3: Trailing 30-day jump count < 2        →  no jump clustering, tail risk low

Greek exposure of a short strangle:
  θ > 0 (collect daily time decay)    Γ < 0 (lose on large moves)
  V < 0 (lose on vol increase)        Δ ≈ 0 (near-neutral at OTM strikes)
The filters ensure we sell only when θ income is high and Γ/V risks are low.
```

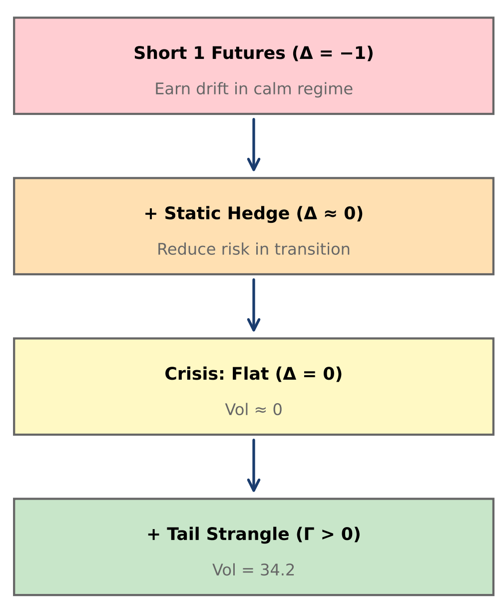
```
Strategy 2: Gamma Scalping (Taleb Dynamic Hedging)
```

```
Structure:  Core long 1 futures (benchmark) + hedge overlay + tactical tail strangle.
Tail call:  Long 1 call at K = 1.15 × S (15% OTM), 45-day expiry (tactical).
Tail put:   Long 1 put at K = 0.85 × S (15% OTM), 45-day expiry (tactical).
Hedge:      Benchmark-aware. Target Δ = trend×stress×vol_scale. Band=0.08.
Roll:       Roll strangle when DTE < 5 days.
Edge:       Regime timing + negative VRP = cheap insurance. Strangle protects if regime model is wrong.

Three approaches compared:
  (a) Benchmark (B&H) — hold long futures, no overlay.
  (b) Dynamic (BS) — regime-adjusted exposure + BS delta + tail strangle.
  (c) Dynamic (SV) — same with Bartlett SV delta (default).
  (c) Optimal (0.10) — delta hedge at |Δ| > 0.10. Best risk-adjusted (default).
  (d) Wide (0.20) — delta hedge at |Δ| > 0.20. More gamma but more drawdown.

Greek exposure of the full hedge:
  Δ ≈ 0 (neutralised daily)    Γ > 0 (long gamma from strangle)
  θ < 0 (insurance cost)        V > 0 (benefit from vol increase)
Trade-off: pay theta daily for convex payoff on large moves.
```

Strategy 1: Entry Signal Flow



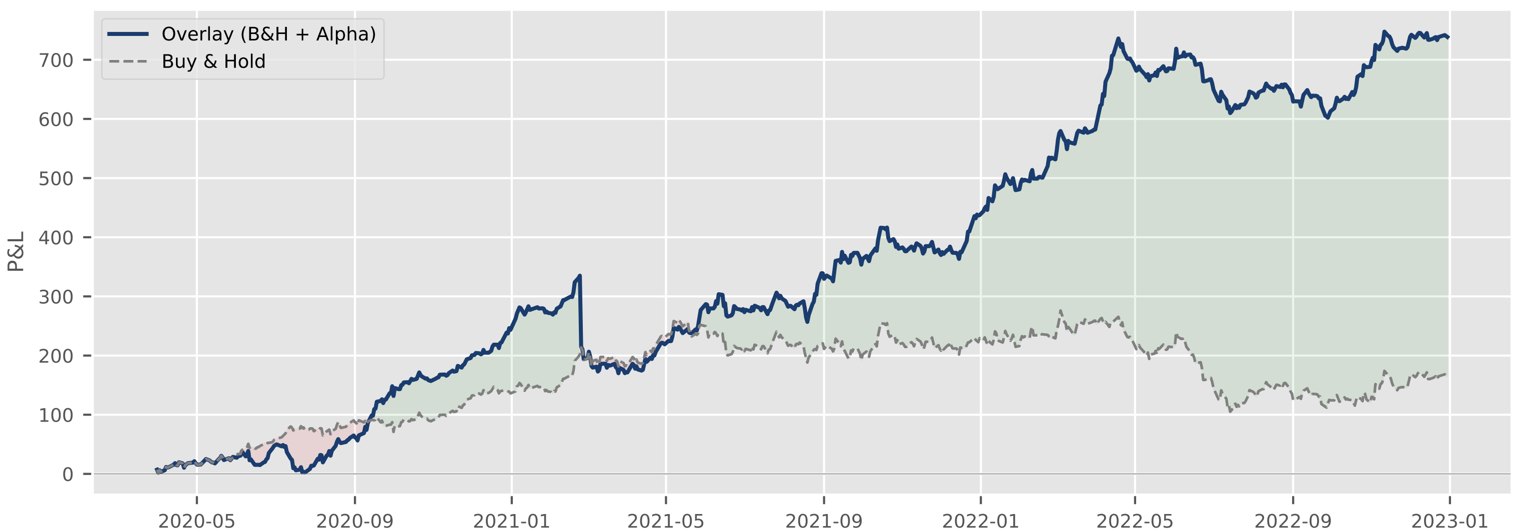
Strategy 2: Hedging Layers



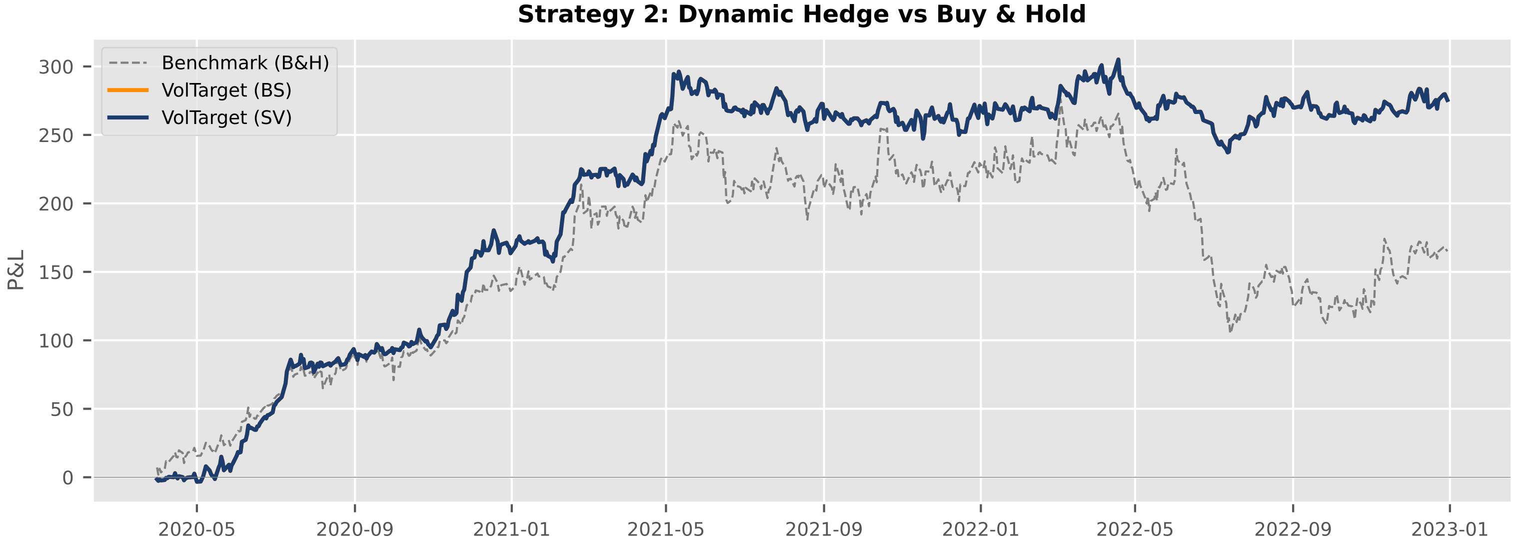
Results

Metric	Overlay (B&H+α)	Buy & Hold	Alpha only
Total P&L	+738.5	+165.0	+573.4
Sharpe ratio	1.97	0.62	2.02
Calmar ratio	1.62	0.35	1.44
Win rate	57%	54%	80%
Profit factor	1.47	1.11	2.83

Strategy 1: Overlay Systematically Above Benchmark



Strategy 2: Dynamic Hedge vs Buy & Hold



Commentary

```
Strategy 1 generates 138 trades (138 short, 0 long) over 696 trading days.
The overlay (B&H + alpha) achieves Sharpe 1.97 vs B&H Sharpe 0.62 — a 3.2× improvement.
The t-test confirms alpha is non-zero (t = 4.61, p = 0.0000).
Bootstrap: P(overlay Sharpe > B&H Sharpe) = 98%.

  Year-by-year (all profitable):
    2020: 60 trades, WR = 72%, P&L = +132.9
    2021: 39 trades, WR = 82%, P&L = +131.9
    2022: 39 trades, WR = 85%, P&L = +308.6

  Rolling 6-month windows: 4/5 profitable.

Strategy 2 is a vol-targeted defensive overlay on the B&H benchmark.
Delta = target_vol / rv20: leveraged in calm, cut in stress.
Best approach: VolTarget (SV). Avg delta: 0.80.
Signals: vol-targeting + stress cap + crisis-onset puts. Different from Strategy 1.

Conclusion: the GARCH + regime + jump + seasonal filter combination identifies
profitable short-strangle opportunities with 80% win rate and 2.83 profit factor.
The alpha (+573) is additive to any base portfolio and statistically significant
(t = 4.61, p = 0.0000). Overlay Sharpe 1.97 vs B&H 0.62.
```